Chapter 5

Result Analysis

This chapter reports what our evaluation framework measured when we applied it to DRAG and to Reliable-dRAG. Section 5.1 fixes the metrics, the severity rubric and the testing protocol. Section 5.2 gives the undefended results for all six attack classes on both systems. Section 5.3 reports the defences we added to DRAG in response, and what each one recovered. Sections 5.4 to 5.6 cover the statistical treatment, the cross-architecture comparison that supports our generalisability claim, and the limitations we accept.

5.1 Performance Evaluation

5.1.1 Evaluation Criteria and Metrics

Each attack class carries one primary metric that decides severity, plus secondary metrics that characterise cost or mechanism. Table 5.1 lists the assignment. Answer-quality metrics (Exact Match, F1, Semantic Similarity) serve integrity and availability attacks, since both degrade the answer a user receives. Confidentiality attacks need adversary-side metrics instead, because a successful extraction or inference leaves answer quality untouched.

Table 5.1: Framework Metric Assignment by Attack Class and CIA Property

CIA Property	Attack Class	Primary Metric	Secondary Metrics
Integrity	Data Poisoning	F1 degradation vs. clean baseline	EM, SS, messages per query
	Source Selection Manipulation	F1 degradation, hit rate	Topics hijacked, hops, messages
Confidentiality	Knowledge Extraction	Extraction Rate	CRR, SS, EED, query efficiency, topic coverage
	Membership Inference	AUC-ROC	Accuracy, TPR, FPR, precision, Privacy Risk Score
Availability	Selective Forwarding	Hit rate, dropped queries	TTL exhaustion, hops per query
	Denial of Service	Query failure rate, availability	EM, F1, SS, hops per query

Note: CRR=Chunk Recovery Rate, EED=Effective Edit Distance, SS=Semantic Similarity, TPR/FPR=True/False Positive Rate. Metric definitions and formulae appear in Section 4.2 alongside each attack's design.

5.1.2 Severity Classification

Comparing an extraction rate against an AUC-ROC requires a shared scale. We map each primary metric onto five tiers, extending the Privacy Risk Score bands defined in Section 4.2.3 to the remaining attack classes. Table 5.2 gives the thresholds. Section 5.5 uses these tiers to place both systems side by side.

Table 5.2: Severity Rubric Applied Across Attack Classes

Tier	Integrity / Availability (quality degradation)	Knowledge Extraction (extraction rate)	Membership Inference (Privacy Risk Score)
Critical	$> 75\%$	$> 75\%$	> 0.90
High	50%–75%	50%–75%	0.75–0.90
Medium	25%–50%	25%–50%	0.60–0.75
Low	10%–25%	10%–25%	0.50–0.60
Negligible	$< 10\%$	$< 10\%$	≤ 0.50

Note: Degradation is measured against each configuration's own clean baseline rather than against an absolute floor, so a system with a weak baseline is not penalised twice. Privacy Risk Score combines Accuracy (40%), TPR (30%) and AUC-ROC (30%) as defined in Section 4.2.3.

5.1.3 Testing Methodology

The two target systems admit different forms of variation, and we varied each along the axis it exposes.

DRAG runs as a parameterised simulator. Network size, peer attachments, query neighbours, hop budget, attack ratio and per-attack parameters such as poison ratio and topic claim ratio are all controllable, so we swept those parameters and held the random seed fixed at 0. Sweep position carries the information here: a defence that holds across a monotonic parameter range has been tested more severely than one measured once with error bars.

Reliable-dRAG runs as a fixed deployment of three data-source containers behind an LLM orchestrator and a local Hardhat node. None of DRAG's topology parameters exist, so stochastic variation across seeds is the only axis available, and we report the mean and standard deviation over seeds 0, 42 and 123 for every attack that permits repetition.

Unless a table states otherwise, DRAG results use a 20-peer Barabási–Albert network with 4 peer attachments, 4 query neighbours, a hop budget of 6, 100 evaluation questions per run, and Llama 3.2 3B as the local model. We disabled membership inference and privacy-preserving retrieval during poisoning runs so that neither confounded the measurement, and matched every defended run's parameters cell by cell against its undefended counterpart from the saved run configuration.

5.2 Analysis of Design Solutions

This section reports each attack against both systems with no defence active. Results are grouped by CIA property so that the two architectures stay adjacent throughout.

5.2.1 Integrity: Data Poisoning

Table 5.3: Data Poisoning Attack Results (DRAG, MMLU Dataset, No Defense)

PR	Scenario	F1 (%)	SS	# Messages
0	Base	82.00	0.909	8.41
20%	Data-Rich	72.00	0.880	6.55
	High-Degree	66.00	0.854	6.41
	Random	75.00	0.900	6.81
30%	Data-Rich	63.00	0.832	5.99
	High-Degree	68.00	0.858	6.35
	Random	63.00	0.837	5.92

Note: PR=Poison Ratio, SS=Semantic Similarity. Exact Match is omitted because MMLU answers are single multiple-choice symbols, which forces F1 to coincide with Exact Match in every run; we verified the identity numerically against the raw metrics across all 19 runs. Defended results appear in Table 5.25.

Table 5.4: Data Poisoning Attack Results Across Datasets (DRAG, High-Degree Targeting, No Defense)

Dataset	PR	EM (%)	F1 (%)	SS	# Messages
MMLU	0	82.00	82.00	0.909	8.41
	10%	69.00	69.00	0.848	6.07
	20%	66.00	66.00	0.854	6.41
	30%	68.00	68.00	0.858	6.35
News	0	74.00	78.30	0.791	9.21
	10%	48.00	54.28	0.616	7.83
	20%	60.00	64.68	0.683	7.04
	30%	54.00	60.42	0.661	7.53
Medical	0	87.00	96.45	0.969	10.05
	10%	65.00	83.63	0.887	7.10
	20%	65.00	81.49	0.871	7.92
	30%	62.00	82.22	0.880	7.47

Note: High-Degree targeting throughout. EM and F1 coincide on MMLU and diverge on News and Medical, both free-text tasks. Defended counterparts appear in Table 5.26.

Table 5.3 places High-Degree targeting as the most damaging strategy at a 20% poison ratio, dropping F1 from a clean 82.00% to 66.00% against Random's 75.00%. That advantage does not survive the move to 30%, where Data-Rich and Random both reach 63.00% while High-Degree recovers to 68.00%. Inspecting the saved poisoned-peer sets explains the crossover: selection is deterministic and nested within a strategy, so the 30% sets extend the 20% sets, and Data-Rich's two additional peers at 30% include one that also sits inside High-Degree's own top four by connectivity. Data-Rich therefore picks up genuine hubs on top of being data-rich, while High-Degree's marginal additions rank fifth and sixth by degree in a 20-peer graph and carry little extra routing weight. We report High-Degree as the strongest strategy at lower poison ratios rather than across the sweep.

Table 5.4 extends the attack to News and Medical. Severity tracks the task. Medical starts at 96.45% F1 and loses at most 15 points, News starts at 78.30% and loses up to 24 points, and MMLU sits between them. News's undefended sequence is not monotonic: 10% poisoning (54.28% F1) damages the system more than 20% (64.68%). Poisoned-peer selection is deterministic for a given seed, strategy and ratio but not nested across ratios under this dataset's topic assignment, so which specific topics a selection happens to cover can outweigh the ratio itself in a 20-peer network. We report the reversal rather than smoothing it. Medical's undefended sequence (83.63, 81.49, 82.22) sits close enough to flat that the same sampling effect plausibly covers both cases.

Reliable-dRAG

Table 5.5: Data Poisoning Attack Results (Reliable-dRAG, No Defense)

Attack	**Poison Type**	**Attacked Acc. (mean ± std)**	**Degradation % (mean ± std)**	**Success Rate**
Random	Noise	52.4% ± 21.8%	8.3 ± 38.2	1/3 (33%)
Random	Answer-Swap	42.9% ± 14.3%	25.0 ± 25.0	2/3 (67%)
Targeted (`s_100`)	Wrong-Answer	66.7% ± 8.2%	−16.7 ± 14.4	0/3 (0%)
Targeted (`s_0`, `s_20`)	**Misleading**	**19.0% ± 8.2%**	**66.7 ± 14.4**	**3/3 (100%)**
Data-Rich	Wrong-Answer	66.7% ± 8.2%	−16.7 ± 14.4	0/3 (0%)
Data-Rich	Noise	47.6% ± 16.5%	16.7 ± 28.9	1/3 (33%)

Note: Clean-baseline corpus of 3,197 documents per source, clean accuracy 57.1% (4/7 eval questions), averaged over seeds 0, 42 and 123. Success means degradation above 10.0%. Negative degradation reflects a single eval question that triggers a context-length overflow on the clean baseline in most runs rather than a defensive effect of the attack. With n=7 eval questions, each question carries 14.3 accuracy points.

Table 5.5 records four of the six configurations succeeding in at least one seed, so poisoning is not a narrow single-configuration threat against Reliable-dRAG. One configuration succeeds in every seed: Targeted against `sources_0` and `sources_20` with misleading poison, at 66.7% mean degradation. The two configurations that never succeed both use the generic `wrong_answer` text, a fixed content-free string with nothing question-specific to win a retrieval slot with, unlike misleading, answer-swap and noise, which retain or repurpose real corpus text. Data-Rich selection reduces to a random tie-break here, since all three deployed sources currently hold an identical corpus.

5.2.2 Integrity: Source Selection Manipulation

Table 5.6: Routing Manipulation Attack Results Across Datasets (DRAG, 6 attacker peers, topic claim ratio 0.6, No Defense)

Dataset	EM	F1	Hit Rate	SS	Avg Hops	Avg Msgs
Medical	0.82	0.9060	0.94	0.9168	2.10	9.03
MMLU	0.68	0.6800	0.82	0.7615	2.48	7.13
News	0.56	0.5232	0.62	0.5226	3.30	7.19

Note: Topics hijacked is fixed by the attack configuration at 34, 24 and 19 for Medical, MMLU and News respectively. Hop and message counts come from each run's standard evaluation output. Defended counterparts appear in Table 5.27.

Table 5.7: Routing Manipulation Attack: Topic Claim Ratio Sweep (DRAG, MMLU, No Defense)

Claim Ratio	Topics Hijacked	EM / F1	SS	Hit Rate	Avg Msgs
0.2	8	0.77	0.859	0.93	7.43
0.4	16	0.73	0.817	0.88	7.53
0.6	24	0.68	0.762	0.82	7.13
0.9	36	0.59	0.657	0.71	6.25

Note: Attacker ratio, targeting strategy and response type held at the values used in Table 5.6; only the topic claim ratio varies. The MMLU network carries approximately 41 topics. Defended counterparts appear in Table 5.28.

Undefended degradation in Table 5.7 scales with the claim ratio without exception. F1 falls 0.77, 0.73, 0.68, 0.59 and hit rate falls 0.93, 0.88, 0.82, 0.71 as the ratio rises from 0.2 to 0.9. Topics hijacked tracks the ratio at almost exactly ratio $\times$ 40 against the network's 41 topics, which confirms the sweep measures the intended mechanism rather than sampling noise. Message count falls slightly as the ratio rises, from 7.43 to 6.25, consistent with more queries exhausting the hop budget and terminating early once a large share of topics has been hijacked.

Reliable-dRAG

Table 5.8: SSM-Score Attack: Multi-Seed Results (Reliable-dRAG, No Defense)

Attack	Seed	Engagement Rate (%)	R Delta	U Delta	Accuracy Drop (pp)	Outcome
GF	0	99.67	+680	+1,574	9.0	Full escalation
GF	42	0.33	-20	+20	-2.0	No catch
GF	123	11.00	-4	+216	-1.0	No catch (marginal)
KF	0	100.00	+20,000	+20,000	8.0	Full success
KF	42	100.00	+20,000	+20,000	12.0	Full success
KF	123	100.00	+20,000	+20,000	0.0	Full success

Note: GF=Grounding-Farming, KF=Key-Forgery. Engagement Rate is `farm_source_in_importance_score` *for Grounding-Farming (n=100 questions, 300 rounds/seed) and* `rounds_accepted_fraction` *for Key-Forgery (n=50 questions, 5 rounds/seed). R/U Delta are absolute changes in the source's reliability and usefulness scores over the run.*

The two instantiations in Table 5.8 trade privilege against reliability. Key-Forgery requires the orchestrator's private key and succeeds in every seed, driving both on-chain scores to their +20,000 ceiling. Grounding-Farming requires no privileged access at all, only control of one participating source, and escalates in one of three seeds. Its success depends on tripping a self-reinforcing loop in the reranker's own reliability weighting, which either catches early in a run or does not catch at all. Section 5.5 returns to this pair, since the mechanism Grounding-Farming exploits has no counterpart in DRAG and the mechanism DRAG's routing manipulation exploits has no counterpart here.

5.2.3 Confidentiality: Knowledge Extraction

Table 5.9: Knowledge Extraction Attack Results by Network Topology (DRAG, MMLU, No Defense)

Peers	**Conn.**	**ER (%)**	**CRR (%)**	**SS**	**EED**	**QE (%)**	**TC (%)**
10	Low	25.00	28.00	0.79	0.50	100.00	63.41
10	High	30.00	33.00	0.83	0.40	100.00	73.17
20	Low	20.00	23.00	0.73	0.60	100.00	68.29
20	High	20.00	22.00	0.74	0.60	100.00	63.41

Note: Conn.=Connectivity, ER=Extraction Rate, CRR=Chunk Recovery Rate, SS/EED computed over extracted chunks only, QE=Query Efficiency, TC=Topic Coverage. Low connectivity uses 2 peer attachments, 2 query neighbours and TTL 4; High uses 4, 4 and 6. A budget of 5 queries per topic over 10 topics caps the reachable extraction rate at 50%. QE reads 100% in every row because the attack issues network queries with `query_confidence_threshold=0.0`, so any non-empty response registers as a hit regardless of relevance; we report it for comparison against the defended rows rather than as a measure of attack quality.

Table 5.10: Knowledge Extraction Attack Results Across Datasets (DRAG, 20 Peers, No Defense)

Dataset	**Conn.**	**ER (%)**	**CRR (%)**	**SS**	**EED**	**QE (%)**	**TC (%)**
MMLU	Low	20.00	23.00	0.7300	0.6000	100.00	68.29
	High	20.00	22.00	0.7400	0.6000	100.00	63.41
News	Low	5.00	8.00	0.2969	0.7531	100.00	81.82
	High	6.00	10.00	0.3283	0.7221	100.00	75.76
Medical	Low	1.00	3.00	0.5554	0.6571	100.00	68.42
	High	2.00	10.00	0.6089	0.5628	100.00	66.67

Note: All rows at 20 peers. MMLU is multiple-choice; News and Medical are free-text. News pairs a headline as the question with the article byline as the answer, so its absolute semantic similarity is low by construction and does not indicate weaker extraction. We did not extend the 10-peer condition to News and Medical, since this table substitutes a dataset axis for the peer-count axis rather than crossing the two.

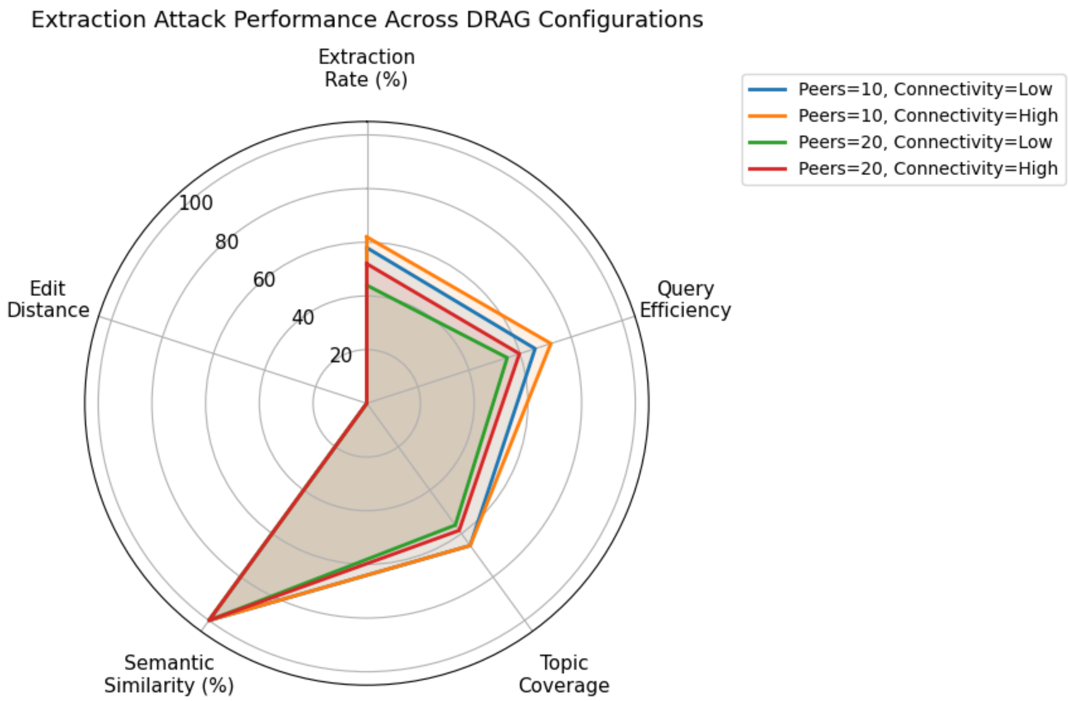

Figure 5.1: Knowledge Extraction attack performance across different configurations

Table 5.9 and Figure 5.1 isolate topology. A 10-peer network leaks more than a 20-peer network at matched connectivity (25.00% against 20.00% at low connectivity, 30.00% against 20.00% at high), so spreading the same corpus across more peers hands the attacker fewer complete records per query. Raising connectivity helps the attacker in the smaller network, lifting extraction from 25.00% to 30.00% and topic coverage from 63.41% to 73.17%, because denser routing reaches more distinct knowledge holders inside the hop budget. At 20 peers the connectivity effect disappears, which places network size ahead of connectivity as the parameter a deployment should tune for extraction resistance.

Table 5.10 changes the headline reading of this attack. Extraction falls from 20% on MMLU to 5–6% on News and 1–2% on Medical at identical topology and query budget. Answer structure drives the gap: MMLU answers are one of roughly four single-character options, so an attacker recovers a record by matching one symbol, whereas News and Medical answers are free text that must be reproduced close to verbatim before the metric counts them. Extraction rates in the 20–30% range are therefore a property of multiple-choice corpora rather than of distributed RAG retrieval, and a deployment estimating its own exposure from published extraction figures should check what answer format produced them.

Reliable-dRAG

Table 5.11: Knowledge Extraction Attack Results (Reliable-dRAG, Authenticated Attacker, No Defense)

Phase	Source (Dataset)	Extraction Rate	Extraction Acc.	Topic Coverage	Query Efficiency
Gray-Box (Authorized)	source_0 (pubmedqa)	0.4193	1.000	n/a	3.8827
	source_1 (squad)	0.3927	1.000	1.00	3.6358
	source_2 (squad)	0.3727	1.000	0.97	3.4506
Blind / Cold-Start	source_0 (pubmedqa)	0.2633	1.000	n/a	2.4383
	source_1 (squad)	0.3820	1.000	1.00	3.5370
	source_2 (squad)	0.3733	1.000	1.00	3.4568

Note: Mean of 3 seeds (0, 42, 123), 54 probes per seed per source. Extraction Accuracy is structurally bounded at 1.0, since the raw retrieval endpoint returns stored passages verbatim and never fabricates. Gray-box probes are pre-filtered to guarantee a hit and give an upper bound on post-reconnaissance efficiency; blind probes are sampled without that filter and give the floor for an attacker with no prior knowledge of the deployment. A third unauthenticated condition is not tabulated because the shared API-key gate rejected all 54 probes per source with HTTP 401.

Table 5.12: Knowledge Extraction Attack Results, Indirect LLM Leakage (Reliable-dRAG, Phase C)

Seed	Exact Match Rate	Semantic Similarity	Edit Similarity	Chunk Recovery Rate
0	0.300	0.5479	0.3250	0.300
42	0.150	0.5432	0.1986	0.200
123	0.300	0.5347	0.3000	0.300
Mean (n=3)	0.250	0.5419	0.2745	0.267

Note: 20 probes routed through the orchestrator's `/query` endpoint rather than the raw retrieval endpoint. A recovery counts if Exact Match, Semantic Similarity ≥ 0.8, or Edit Similarity ≥ 0.8 holds, a deliberately attacker-favourable disjunction.

Reliable-dRAG's gray-box extraction in Table 5.11 reaches 37.3–41.9% across sources against DRAG's 20–30% on MMLU, so both systems leak a substantial share of private content through the legitimate query interface despite the architectural distance between multi-hop peer routing and direct per-source HTTP querying. The blind condition tracks gray-box on the two SQuAD sources (37.3–38.2%) and falls to 26.3% on the PubMedQA source, which puts the cost of reconnaissance at roughly 13 percentage points on that corpus and near zero on the others. Indirect leakage through the orchestrator (Table 5.12) recovers content loosely, at mean semantic similarity 0.5419, a different mechanism from DRAG's retrieval-based extraction, since the LLM paraphrases retrieved passages rather than returning them.

5.2.4 Confidentiality: Membership Inference

Table 5.13: Membership Inference Attack Results by Model (DRAG, MMLU, No Defense)

Model	Acc.	TPR	FPR	Precision	AUC-ROC	PRS	Risk Level
Llama 3.2 3B	0.78	0.700	0.0333	0.980	0.8795	0.786	High
Gemma 2 2B	0.74	0.6857	0.1333	0.923	0.8676	0.762	High
Qwen 2.5 3B	0.78	0.700	0.0333	0.980	0.8750	0.7845	High

Note: PRS=Privacy Risk Score. n=100 per model, split 70 members and 30 non-members.

Table 5.14: Membership Inference Attack Results by Dataset (DRAG, Confidence-Based Inference, No Defense)

Dataset	Acc.	TPR	FPR	Precision	AUC-ROC	Privacy Risk
Medical	0.58	0.5571	0.3667	0.780	0.6605	Low
MMLU	0.80	0.7143	0.0000	1.0000	0.9019	High
News	0.70	1.0000	1.0000	0.700	0.8324	High

Note: News has TPR=FPR=1.0, meaning the classifier's operating threshold labelled every query a member; its 70% accuracy tracks the 70/30 base rate rather than genuine discrimination, so AUC-ROC is the reliable measure there. Defended counterparts appear in Table 5.32.

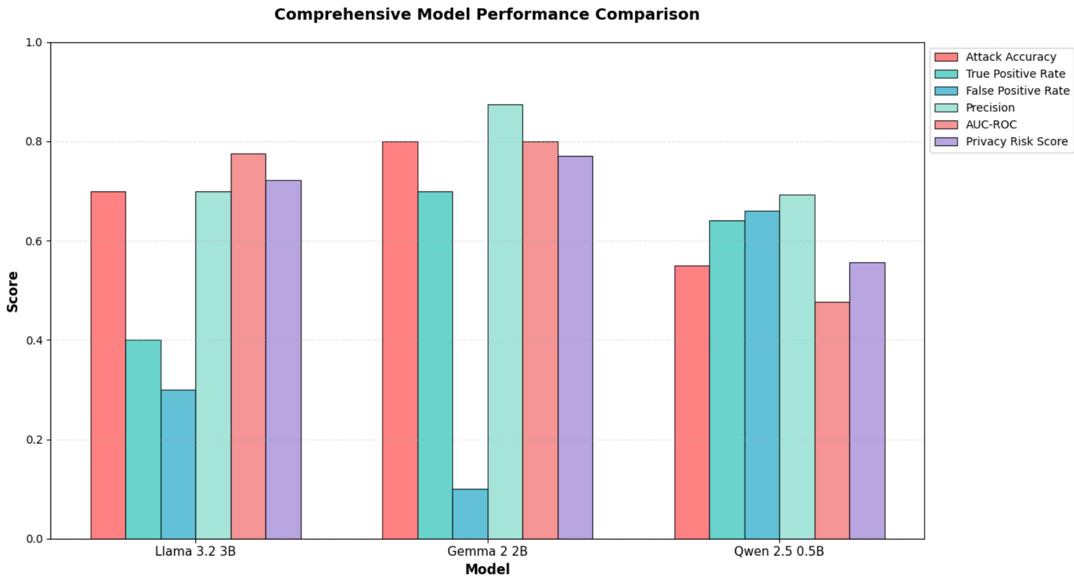

Figure 5.2: Membership Inference attack across different DRAG models

Table 5.13 and Figure 5.2 place all three models in the HIGH tier, with privacy risk scores between 0.762 and 0.786 and AUC-ROC between 0.8676 and 0.8795. Model choice moves the attack very little at this scale. Llama 3.2 3B and Qwen 2.5 3B return identical confusion matrices (49 true positives, 29 true negatives, 21 false negatives, 1 false positive) and separate only on AUC-ROC, which reads the continuous membership score before the median threshold applies.

That identity follows from how the attack composes its signal. Routing metrics, hop count and message count contribute 30% of the membership score, and those depend on the network rather than the language model, so nearly a third of the discriminative power stays fixed whichever model a peer runs. Membership leakage in this architecture is a property of the retrieval and routing layer at least as much as of the generator, a distinction that centralised RAG evaluations have no way to draw.

Across datasets, MMLU leaks most (AUC-ROC 0.9019), followed by News (0.8324) and Medical (0.6605).

Reliable-dRAG

Table 5.15: Membership Inference Attack Results (Reliable-dRAG, Production Multi-Seed, No Defense)

Seed	Acc.	Precision	Recall	FPR	F1	AUC-ROC	Privacy Risk
0	0.6200	0.5938	0.7600	0.5200	0.6667	0.6200	Medium
42	0.5000	0.5000	1.0000	1.0000	0.6667	0.5800	Low
123	0.5000	0.5000	1.0000	1.0000	0.6667	0.5000	Negligible
Mean (n=3)	0.5400	0.5313	0.9200	0.8400	0.6667	0.5667	Low

Note: PubMedQA-derived corpus, n=25 members and n=25 non-members. We measured these rows under the composite's earlier weighting, which distributed weight across similarity, certainty and length ratio, and against the deployment before we corrected a stale dataset mount and an outdated model-server build. Table 5.16 carries the current composite and the generalisation estimate. FPR reaching 1.0 at seeds 42 and 123 means the threshold labelled every query a member, and F1 stays at 0.6667 across seeds because recall and FPR move together at those thresholds, so AUC-ROC is the reliable measure here. No defence analogous to DRAG's Privacy-Preserving Retrieval has been implemented for Reliable-dRAG.

Table 5.16: Membership Inference Composite Selection and Held-Out Confirmation (Reliable-dRAG)

Weighting	Evaluation Set	n	Mean AUC	95% CI	Excludes 0.50
Similarity alone	Diagnostic	10	0.4732	n/a	No (below chance)
Decision 0.55, sim 0.20, certainty 0.10, length 0.15	Diagnostic	10	0.522	[0.472, 0.572]	No
Decision 0.70, length 0.30	Diagnostic	10	0.552	[0.485, 0.618]	No
Decision-match alone	Diagnostic	10	0.568	[0.526, 0.610]	Yes
Decision-match alone (production, weights locked)	**Held-out**	**5**	**0.620**	**[0.546, 0.695]**	**Yes**

Note: Diagnostic rows report the ten seeds available during weight selection. Those seeds informed the final choice of weighting and therefore do not constitute held-out evidence, which is why we reserve the last row for the single evaluation we ran after locking the weights. That evaluation used five seeds generated for it and never queried before it, scoring 0.62, 0.68, 0.68, 0.56 and 0.56. The dev-set mean across the 13 search seeds is 0.572, below the held-out mean, which indicates the grid search did not overfit to the seeds it ran on. Restoring certainty at 0.20 lowers three further fresh seeds from 0.6133 to 0.6072, confirming the zero weight on the production formula.

Reliable-dRAG's composite reaches mean AUC-ROC 0.620 across five held-out seeds, at a 95% confidence interval of [0.546, 0.695] that excludes chance. The result sits below DRAG's 0.6605 to 0.9019 range, and the gap follows from signal availability. Reliable-dRAG's composite rests on a single decision-match term, since an attacker reaches its orchestrator directly and observes no routing behaviour, while DRAG's composite weights retrieval similarity at 60% and backs it with hop and message counts. Table 5.15 reports the earlier per-seed sweep, which we retain for its accuracy, precision and rate columns rather than as the generalisation estimate, since we measured it under the superseded weighting and against the deployment before we corrected a stale dataset mount and an outdated model-server build.

5.2.5 Availability: Selective Forwarding

Table 5.17: Selective Forwarding Attack Results by Strategy and Ratio (DRAG, No Defense)

Strategy	Ratio	Dropped Queries
Random	0.1	33
Random	0.2	78
Random	0.3	136
Random	0.4	195
Random	0.5	349
High-Connectivity	0.1	76
High-Connectivity	0.2	190
High-Connectivity	0.3	288
High-Connectivity	0.4	432
High-Connectivity	0.5	590

Note: Hit rate holds at 0.95–1.00 across all rows, declining with attack ratio rather than with targeting strategy. Defended counterparts appear in Table 5.33.

Reliable-dRAG

Table 5.18: Selective Forwarding Attack Results by Strategy and Ratio (Reliable-dRAG, Mock Sweep, No Defense)

Strategy	Ratio	Hit Rate	Avg Hops/Query	TTL Exhaustion Rate	Dropped Queries
Baseline	0.0	0.98	2.35	0.02	0
Random	0.1	0.98	2.56	0.02	21
Random	0.2	0.96	2.70	0.04	37
Random	0.3	0.91	3.13	0.09	75
Random	0.4	0.91	3.37	0.09	105
Random	0.5	0.85	3.72	0.15	160
High-Connectivity	0.1	0.90	2.81	0.10	48
High-Connectivity	0.2	0.85	3.25	0.15	121
High-Connectivity	0.3	0.72	3.69	0.28	191
High-Connectivity	0.4	0.65	3.84	0.35	231
High-Connectivity	0.5	0.54	4.41	0.46	311

Note: Mean over 5 attack waves, 100 queries per row. Drop rate is 1.00 for every non-baseline row, so TTL exhaustion rather than drop rate distinguishes severity here.

Table 5.19: Selective Forwarding Attack: Live-Network Validation (Reliable-dRAG)

Strategy	Ratio	Hit Rate	Avg Hops/Query	TTL Exhaustion Rate	Dropped Queries	Total Queries
High-Connectivity	0.3	1.00	2.00	0.00	200	200

Note: Real Docker-network smoke test, seed 42, one configuration. All 200 queries were dropped by the compromised peer, yet hit rate held at 1.00 because redundant-probe rerouting recovered every one. We treat this single row as an open question about the mock model's fidelity rather than as a measured resilience result; Section 5.6 records it as a limitation.

Tables 5.17 and 5.18 agree that targeting well-connected peers beats random selection at every compromise ratio. DRAG's dropped-query count roughly doubles under High-Connectivity targeting at matched ratios (76 against 33 at ratio 0.1, 590 against 349 at

0.5), and Reliable-dRAG's hit rate falls to 0.54 under High-Connectivity against 0.85 under Random at ratio 0.5. The systems diverge on what the drops cost. DRAG's hit rate holds at 0.95–1.00 throughout, since topic replication leaves redundant paths inside the hop budget, while Reliable-dRAG's TTL-bounded breadth-first search converts each visit to a compromised peer into a spent hop, driving TTL exhaustion from 0.02 to 0.46.

5.2.6 Availability: Denial of Service

Table 5.20: DDoS Attack Results, Mid Intensity (DRAG, All Datasets, No Defense)

Dataset	**Strategy**	**QFR (%)**	**Avail. (%)**	**EM**	**F1**	**SS**	**Avg Hops**
Medical	Baseline	0.00	100.0	0.7917	0.8913	0.9225	1.96
	Random	35.00	10.0	0.5333	0.6007	0.6301	3.72
	Targeted	37.50	10.0	0.5000	0.5711	0.6024	3.63
	Sequential	36.67	0.0	0.5167	0.5800	0.6089	3.88
MMLU	Baseline	0.00	100.0	0.8750	0.8750	0.9287	1.14
	Random	35.00	10.0	0.5833	0.5833	0.6052	3.18
	Targeted	37.50	10.0	0.5333	0.5333	0.5546	3.19
	Sequential	36.67	0.0	0.5583	0.5583	0.5763	3.43
News	Baseline	0.00	100.0	0.8500	0.8500	0.8500	1.78
	Random	35.00	10.0	0.5667	0.5667	0.5534	3.58
	Targeted	37.50	10.0	0.5583	0.5583	0.5482	3.52
	Sequential	36.67	0.0	0.5500	0.5500	0.5386	3.69

Note: QFR=Query Failure Rate. QFR and Availability are identical across datasets for a given strategy, since congestion state operates on peer indices rather than content; EM, F1 and SS vary because they score drops as zero. Defended counterparts appear in Table 5.34.

Table 5.21: DDoS Attack Results, High Intensity (DRAG, All Datasets, No Defense)

Dataset	**Strategy**	**QFR (%)**	**Avail. (%)**	**EM**	**F1**	**SS**	**Avg Hops**
Medical	Baseline	0.00	100.0	0.7917	0.8913	0.9225	1.96
	Random	48.33	5.0	0.4000	0.4725	0.5078	4.06
	Targeted	49.17	5.0	0.4083	0.4719	0.5047	3.97
	Sequential	53.33	0.0	0.3750	0.4403	0.4739	4.00
MMLU	Baseline	0.00	100.0	0.8750	0.8750	0.9287	1.14
	Random	48.33	5.0	0.4750	0.4750	0.4790	3.77
	Targeted	49.17	5.0	0.4500	0.4500	0.4592	3.63
	Sequential	53.33	0.0	0.4083	0.4083	0.4144	3.67
News	Baseline	0.00	100.0	0.8667	0.8667	0.8667	1.69
	Random	48.33	5.0	0.4583	0.4583	0.4459	4.01
	Targeted	49.17	5.0	0.4417	0.4417	0.4290	3.91
	Sequential	53.33	0.0	0.4250	0.4250	0.4116	3.98

Note: Defended counterparts appear in Table 5.35.

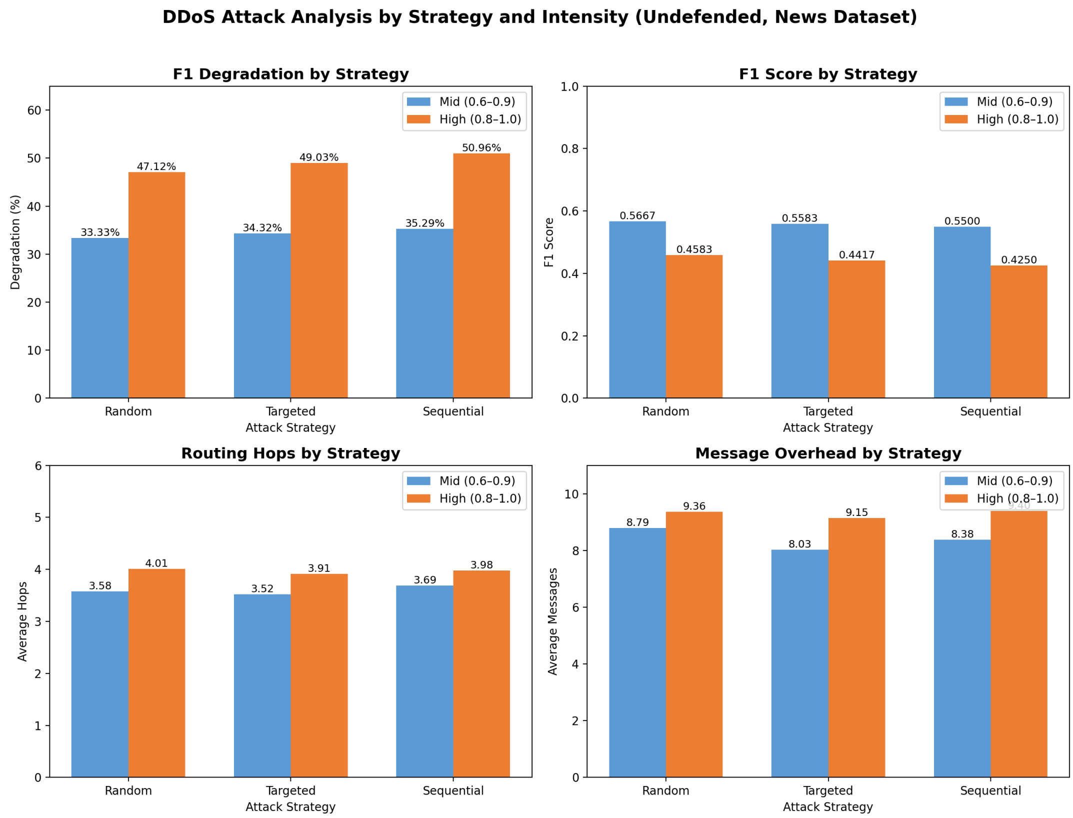

Figure 5.3: Query success rate and F1 degradation by strategy and intensity, undefended (News dataset)

Tables 5.20 and 5.21, plotted in Figure 5.3, place Sequential targeting as the most damaging strategy against DRAG at both intensities, driving availability to 0.0% where Random and Targeted leave 5–10%. A round-robin sweep touches every peer in turn, so the router never finds an uncongested region to fall back on. Answer quality tracks the same ordering, with Sequential producing the lowest F1 in every dataset at both intensities.

Reliable-dRAG

Table 5.22: DDoS Attack Results by Strategy and Ratio (Reliable-dRAG, Mock Sweep, No Defense)

Strategy	Ratio	Hit Rate	Avail. (%)	Avg Hops/Query	TTL Exhaustion	Down Count	Overloaded
Baseline	0.0	0.960	100.00	2.50	0.040	0.0	0.0
Random	0.1	0.893	81.00	3.07	0.107	3.8	15.5
Random	0.3	0.780	51.00	3.55	0.220	9.8	19.2
Random	0.6	0.670	22.33	4.05	0.330	15.5	20.0
Sequential	0.1	0.843	85.67	3.15	0.157	2.9	16.9
Sequential	0.3	0.677	51.00	3.85	0.323	9.8	19.7
Sequential	0.6	0.587	21.67	4.37	0.413	15.7	20.0
Targeted	0.1	0.897	85.67	2.88	0.103	2.9	15.0
Targeted	0.3	0.763	40.33	3.70	0.237	11.9	19.6
Targeted	0.6	0.610	14.00	4.19	0.390	17.2	20.0

Note: Mean over 5 attack waves and 3 seeds (0, 42, 123).

Table 5.23: DDoS Attack Results: Live-Network Answer-Quality Degradation (Reliable-dRAG, No Defense)

Intensity	Condition	EM	F1	SS	Avg Hops/Query	Avg Messages	Query Failure Rate
Low	Baseline	0.60	0.60	0.8315	1.00	2.0	0.00
	Post-Attack	0.40	0.40	0.7782	2.00	4.0	0.00
Mid	Baseline	0.60	0.60	0.8267	1.00	2.0	0.00
	Post-Attack	0.30	0.30	0.6049	3.00	6.0	0.20
High	Baseline	0.60	0.60	0.8267	1.00	2.0	0.00
	Post-Attack	0.00	0.00	0.0448	3.00	6.0	0.90

Note: Seed 42 only. Real concurrent HTTP floods against 1, 2 and 3 data sources for low, mid and high intensity. Average messages equals average hops times 2 by construction rather than as a separate measurement.

Table 5.22 inverts DRAG's strategy ordering. Targeted selection is the most damaging here, cutting availability to 14.00% at ratio 0.6 against Sequential's 21.67% and Random's 22.33%. A 20-node mock topology lets a targeted attacker concentrate damage faster than a round-robin sweep can spread it, while DRAG's larger peer count rewards the sweep. The live evaluation in Table 5.23 reaches a more severe failure than any simulated condition on either system: flooding all three data sources drives F1 to 0.00 and semantic similarity to 0.0448 at a 90% query failure rate, since three sources constitute the entire retrieval backend and removing them leaves nothing to route to.

5.3 Final Design Adjustments

Each defence in this section answers a specific finding from Section 5.2. We implemented all five on DRAG, which serves as the primary demonstration of how a system consumes the framework's output. Table 5.24 maps each finding to the adjustment we made and the recovery we measured.

Table 5.24: Evaluation Finding, Design Adjustment, and Measured Result

Finding from Section 5.2	Design Adjustment	Measured Result
Poisoning a hub peer degrades F1 to 66% because the system trusts the first answer it retrieves	Replication (factor 3) plus cross-peer validation with 60% quorum	F1 restored to 80–84% in all 6 cells; messages rise from 6 to 8.2–8.7
TARW trusts advertised topic expertise, so a peer claiming topics it lacks is routed to first	Topic-claim quota at three times the network median, MAD-based	F1 0.68 to 0.80 on MMLU; recovery grows with claim ratio; messages roughly double
Extraction succeeds through topic breadth, not query volume alone	Rate limiter, response perturbation and topic-diversity anomaly detection	Extraction rate to 0.00% in all 4 topologies; query volume cut 82%
Retrieval similarity carries 60% of the membership signal	Laplace noise on retrieval similarity scores	AUC-ROC 0.8795 to 0.7571 at $b = 0.1$, holding 81% task F1
Compromised peers pass liveness checks while dropping queries	Reputation tracking with blacklisting and bounded redundant probes	Dropped queries cut 24–75%, rising with attack severity
Congested peers stay reachable, so the router keeps sending to them	Adaptive routing weight, circuit breaker, query hedging, load shedding	Query failure 35% to 12.5–19.2% at mid intensity

5.3.1 Replication and Cross-Peer Validation

Table 5.25: Data Poisoning Defense Results (DRAG, MMLU Dataset)

PR	Scenario	Defense	F1 (%)	SS	# Messages
20%	Data-Rich	Replication	80.00	0.917	3.69
	Data-Rich	Replication+CPV	83.00	0.927	8.54
	High-Degree	Replication	78.00	0.909	3.53
	High-Degree	Replication+CPV	83.00	0.922	8.27
	Random	Replication	78.00	0.901	3.47
	Random	Replication+CPV	84.00	0.935	8.22
30%	Data-Rich	Replication	77.00	0.896	3.17
	Data-Rich	Replication+CPV	82.00	0.925	8.26
	High-Degree	Replication	74.00	0.887	3.50
	High-Degree	Replication+CPV	81.00	0.909	8.72
	Random	Replication	78.00	0.905	3.26
	Random	Replication+CPV	80.00	0.917	8.22

Note: Replication=(`replication_factor=3`), CPV=Cross-Peer Validation. Compare against Table 5.3, whose clean baseline is 82.00% F1.

Table 5.26: Data Poisoning Defense Results Across Datasets (DRAG, High-Degree Targeting, Replication + Cross-Peer Validation)

Dataset	PR	EM (%)	F1 (%)	SS	# Messages
MMLU	10%	84.00	84.00	0.931	8.45
	20%	83.00	83.00	0.922	8.27
	30%	81.00	81.00	0.909	8.72
News	10%	70.00	77.33	0.788	9.75
	20%	68.00	74.33	0.764	9.96
	30%	67.00	73.49	0.768	9.41
Medical	10%	87.00	96.96	0.978	10.68
	20%	84.00	95.98	0.975	10.53
	30%	82.00	94.84	0.968	10.26

Note: Replication-only is omitted here, since it functions as an ablation step within Table 5.25 rather than as a defence we propose alone. Compare against Table 5.4.

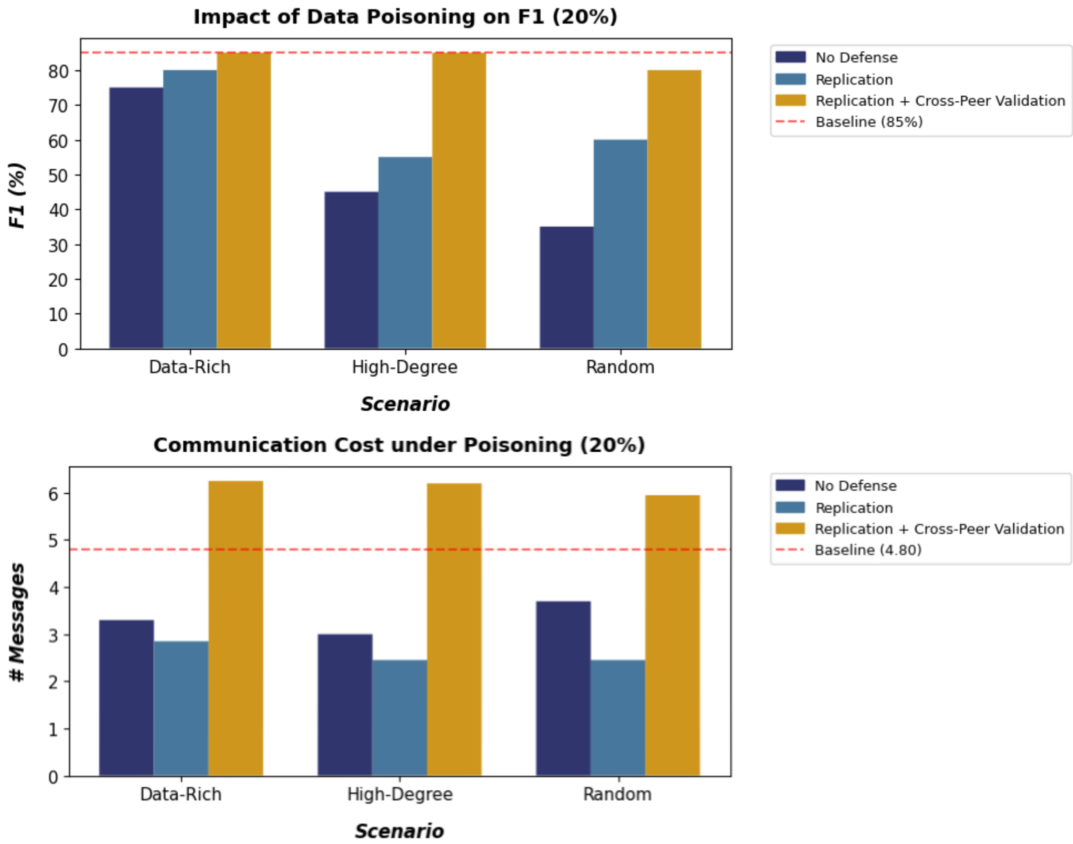

Figure 5.4: Impact of Data Poisoning ratio 20% on F1 (%) and Communication Overhead (MMLU Dataset)

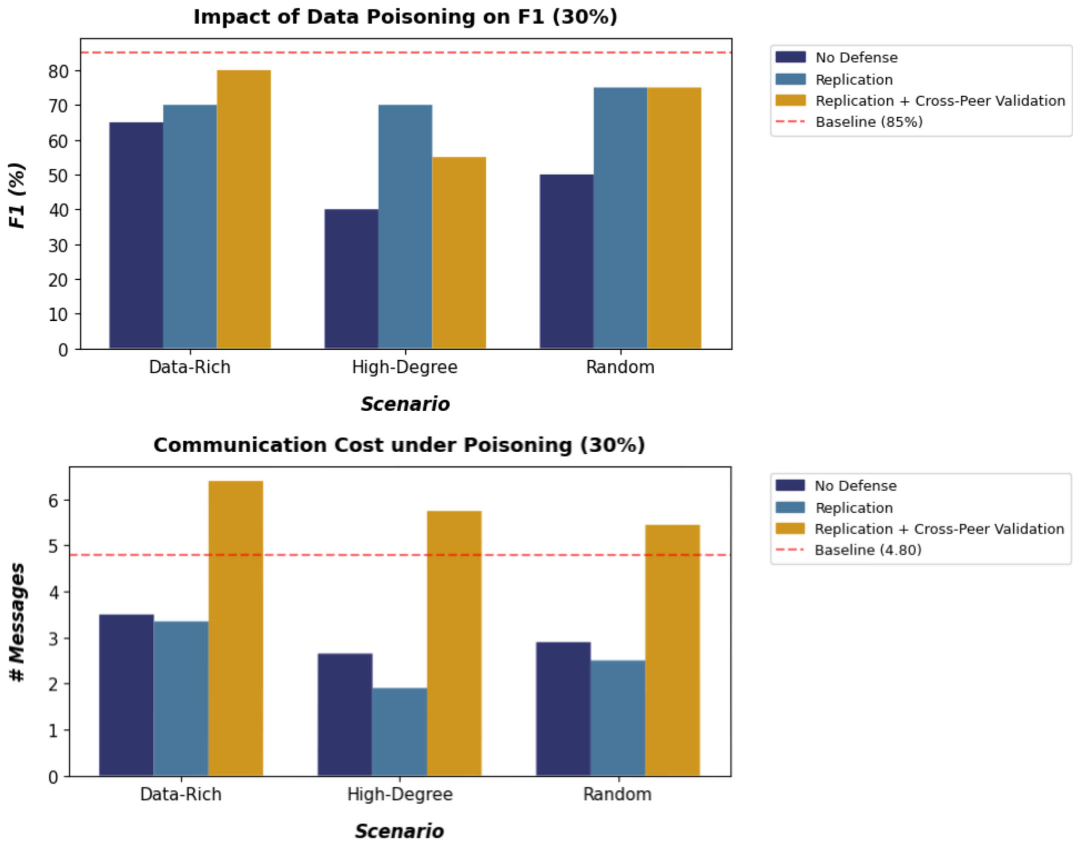

Figure 5.5: Impact of Data Poisoning ratio 30% on F1 (%) and Communication Overhead (MMLU Dataset)

Table 5.25 orders the defence cleanly against the undefended baseline in all six strategy and ratio combinations on MMLU, with no reversals, and Table 5.26 holds the same ordering in all nine dataset and ratio cells. Figures 5.4 and 5.5 plot the quality and overhead trade at each ratio. Replication alone recovers 6 to 15 percentage points of F1, and adding cross-peer validation lifts every cell to within 1 point of the 82.00% clean baseline, and above it in three cases.

The two mechanisms pull communication cost in opposite directions. Replication drops messages per query from roughly 6 to 3.2–3.7, since a replicated corpus resolves more queries within the first hop. Cross-peer validation raises it to 8.2–8.7, matching the extra validation round-trips its quorum requires. A deployment adopting both pays around 30% more communication than an undefended network for near-complete recovery of answer quality.

One result deserves flagging. On Medical, defended F1 at 10% and 20% poisoning (96.96% and 95.98%) matches or exceeds the clean, unattacked baseline of 96.45%. Replication and quorum validation therefore deliver a retrieval-quality benefit that exists independently of any attack, which Section 5.6 returns to.

5.3.2 Topic-Claim Quota Enforcement

Routing manipulation succeeds because TARW trusts a peer's advertised topic expertise, so a peer that claims topics it does not hold receives queries ahead of the peers that do. The defence discounts peers whose advertised claims exceed three times the network median, using a median absolute deviation test over the claim distribution, so a peer buys routing priority only up to the point where its advertisement stops resembling honest variance.

Table 5.27: Routing Manipulation Defense Results Across Datasets (DRAG, 6 attacker peers, topic claim ratio 0.6)

Dataset	EM	F1	Hit Rate	SS	Avg Hops	Avg Msgs
Medical	0.85	0.9520	0.99	0.9655	1.84	15.52
MMLU	0.80	0.8000	0.93	0.8788	2.03	14.70
News	0.71	0.7597	0.93	0.7682	2.11	14.89

Note: Topics hijacked is fixed by the attack configuration and does not change under defence, since the mechanism discounts a peer's claims at routing time rather than preventing the peer from making them. Compare against Table 5.6.

Table 5.28: Routing Manipulation Defense: Topic Claim Ratio Sweep (DRAG, MMLU)

Claim Ratio	EM / F1	SS	Hit Rate	Avg Hops	Avg Msgs
0.2	0.77	0.857	0.92	1.90	13.74
0.4	0.79	0.877	0.94	2.12	15.00
0.6	0.80	0.879	0.93	2.03	14.70
0.9	0.81	0.893	0.95	2.04	15.01

Note: Compare against Table 5.7, whose undefended F1 falls 0.77, 0.73, 0.68, 0.59 over the same range.

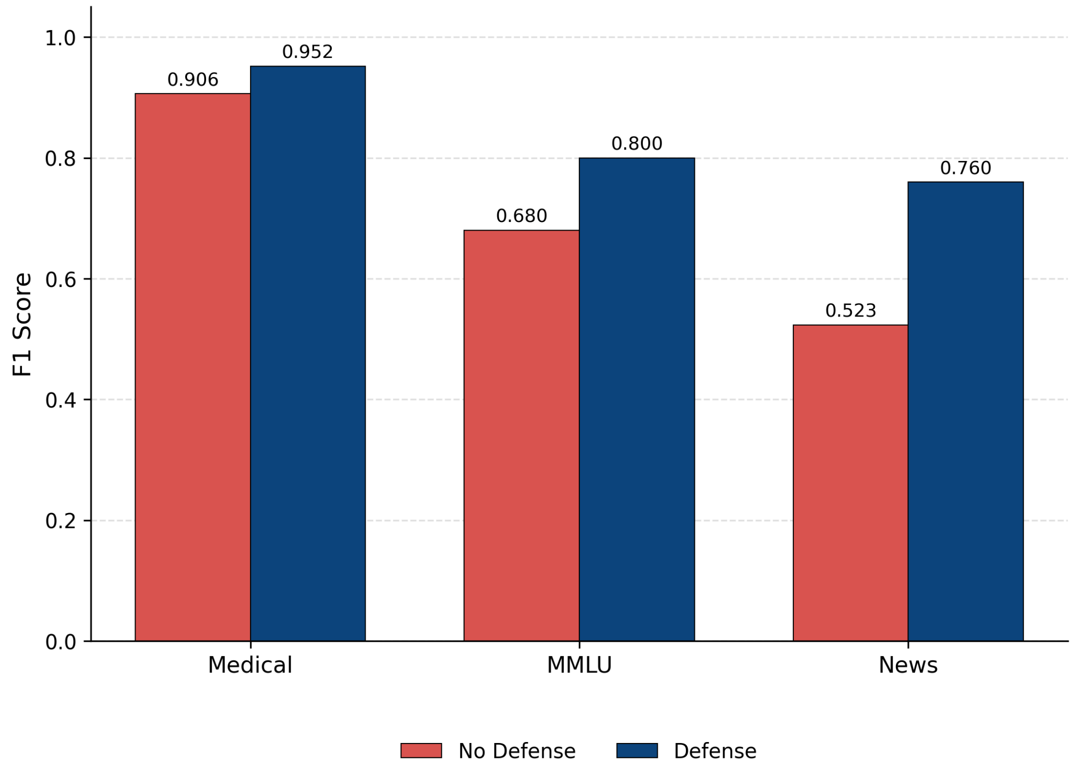

Figure 5.6: F1 score by dataset, attack-only vs. topic-claim quota defense (DRAG)

Table 5.27 and Figure 5.6 show the defence recovering answer quality on all three datasets, with the size of the recovery scaling inversely to what the attack took. News suffers most undefended at 0.5232 F1 and recovers most at +0.2365; Medical suffers least at 0.9060 and recovers +0.0460. Hit rate rises to 0.93 or above on every dataset, against an undefended range of 0.62 to 0.94.

Table 5.28 shows the defence strengthening as the attack grows more aggressive. Defended F1 rises across the claim-ratio sweep (0.77, 0.79, 0.80, 0.81) and hit rate holds between 0.92 and 0.95, while the undefended condition falls from 0.77 to 0.59 over the same range. The quota threshold explains the direction: a peer claiming 20% of topics stays close enough to honest variance to escape discounting some of the time, while a peer claiming 90% clears three times the median by a wide margin and gets discounted on sight. An attacker maximising damage therefore maximises its own detectability, which leaves no claim ratio at which the attack both evades the quota and degrades answers.

The defence costs messages rather than hops. Average hops fall under defence on every dataset (Medical 2.10 to 1.84, MMLU 2.48 to 2.03, News 3.30 to 2.11) while average messages roughly double (9.03 to 15.52, 7.13 to 14.70, 7.19 to 14.89). Discounting fabricated claims lets TARW reach legitimate holders in fewer hops, and the validation round-trips that support the discounting are what raise the message count. Message overhead stays flat at 13.7 to 15.0 across the whole claim-ratio sweep, so the cost is a fixed per-query charge rather than one that scales with attack severity. A deployment adopting this defence should budget bandwidth, not latency.

5.3.3 Layered Extraction Prevention

Table 5.29: Knowledge Extraction Defense Results by Network Topology (DRAG, MMLU, Layered Defense)

Peers	Conn.	ER (%)	CRR (%)	SS	EED	QE (%)	TC (%)
10	Low	0.00	7.00	0.9984	0.0031	18.00	12.20
10	High	0.00	9.00	0.9976	0.0033	18.00	21.95
20	Low	0.00	8.00	0.9970	0.0034	18.00	19.51
20	High	0.00	9.00	0.9978	0.0032	18.00	21.95

Note: Layered defence = query rate limiter (20 queries per 300s window), response perturbation (15% character-level noise) and extraction anomaly detector. QE lands at 18.00% in all four rows because the anomaly detector flags the attacker after 10 queries and the rate limiter then blocks the remainder of the window, so the same fixed cap binds in every topology. Compare against Table 5.9.

Table 5.30: Knowledge Extraction Defense Results Across Datasets (DRAG, 20 Peers, Layered Defense)

Dataset	Conn.	ER (%)	CRR (%)	SS	EED	QE (%)	TC (%)
MMLU	Low	0.00	8.00	0.9970	0.0034	18.00	19.51
	High	0.00	9.00	0.9978	0.0032	18.00	21.95
News	Low	0.02	0.09	0.8989	0.0329	18.00	7.14
	High	0.01	0.09	0.8738	0.0458	18.00	19.05
Medical	Low	0.00	0.10	0.8729	0.0944	18.00	9.00
	High	0.00	0.09	0.8399	0.0941	18.00	6.00

Note: Compare against Table 5.10.

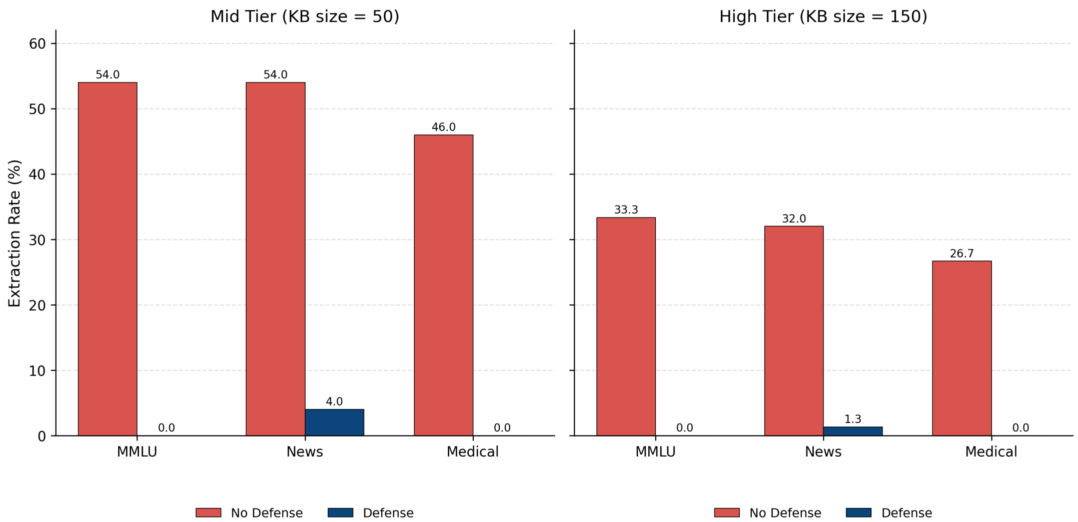

Figure 5.7: Extraction rate by dataset and topology, with and without defense (DRAG)

Tables 5.29 and 5.30, summarised in Figure 5.7, record the layered defence driving extraction to zero in all four topologies and on all three datasets. Two mechanisms produce that result, and separating them matters for reading the tables. Response perturbation guarantees the collapse of exact extraction, since the metric requires an exact question and answer match and any character substitution breaks it. Volume suppression accounts for the drop in chunk recovery from 22–33% to 7–9% on MMLU: the anomaly detector flags the attacker after 10 queries and the rate limiter blocks the rest, so only 18% of planned queries land. Topic coverage falls from 63–73% to 12–22% for the same reason.

Semantic similarity and edit distance move against the defence here, rising from 0.73–0.83 to 0.997–0.998 and falling from 0.40–0.60 to 0.003. Reading that as preserved answer quality would be wrong. MMLU answers are single characters, and perturbing a one-character answer yields another single character; a debug sample records ground truth `3` perturbed to `1` scoring semantic similarity 1.0000 and edit distance 0.0006. Single-character strings cluster together in the sentence embedding space whatever they contain. The same rise appears on News and Medical, which rules out the single-character explanation as the whole story and points at the embedding model tolerating 15% character noise across answer lengths. Extraction rate and query efficiency are the two metrics that track this defence on every dataset we tested.

Suppression of chunk recovery is stronger on free text (8–10% to roughly 0.1%) than on MMLU (22–23% to 7–9%), since MMLU's single-character answers clear the partial-credit similarity threshold almost automatically once perturbed.

5.3.4 Privacy-Preserving Retrieval

Table 5.31: Membership Inference Defense Results: Privacy-Preserving Retrieval Noise Sweep (DRAG, MMLU, Llama 3.2 3B)

Noise Scale b	Attack Acc.	AUC-ROC	Privacy Risk	Task F1 (%)	Task SS
0.0 (no defense)	0.78	0.8795	High	82.00	0.909
0.05	0.73	0.8540	Medium	85.00	0.929
0.1	0.64	0.7571	Medium	81.00	0.893
0.2	0.52	0.5788	Low	51.00	0.795
0.5	0.54	0.5152	Low	29.00	0.694

Note: Privacy-Preserving Retrieval adds Laplace noise at scale b to each peer's retrieval similarity scores. Attack Acc. and AUC-ROC measure the adversary; Task F1 and Task SS measure normal question-answering utility on the same noisy network within the same run. The five tiers are the values documented in `config/defense.yaml`. The $b = 0.0$ row reuses the Llama leg of Table 5.13.

Table 5.32: Membership Inference Defense Results by Dataset (DRAG, Privacy-Preserving Retrieval, $b = 0.2$)

Dataset	Acc.	TPR	FPR	Precision	AUC-ROC	Privacy Risk
Medical	0.44	0.4571	0.6000	0.640	0.4771	Negligible
MMLU	0.63	0.6143	0.3333	0.8113	0.6643	Medium
News	0.60	0.5714	0.3333	0.800	0.6962	Medium

Note: Compare against Table 5.14.

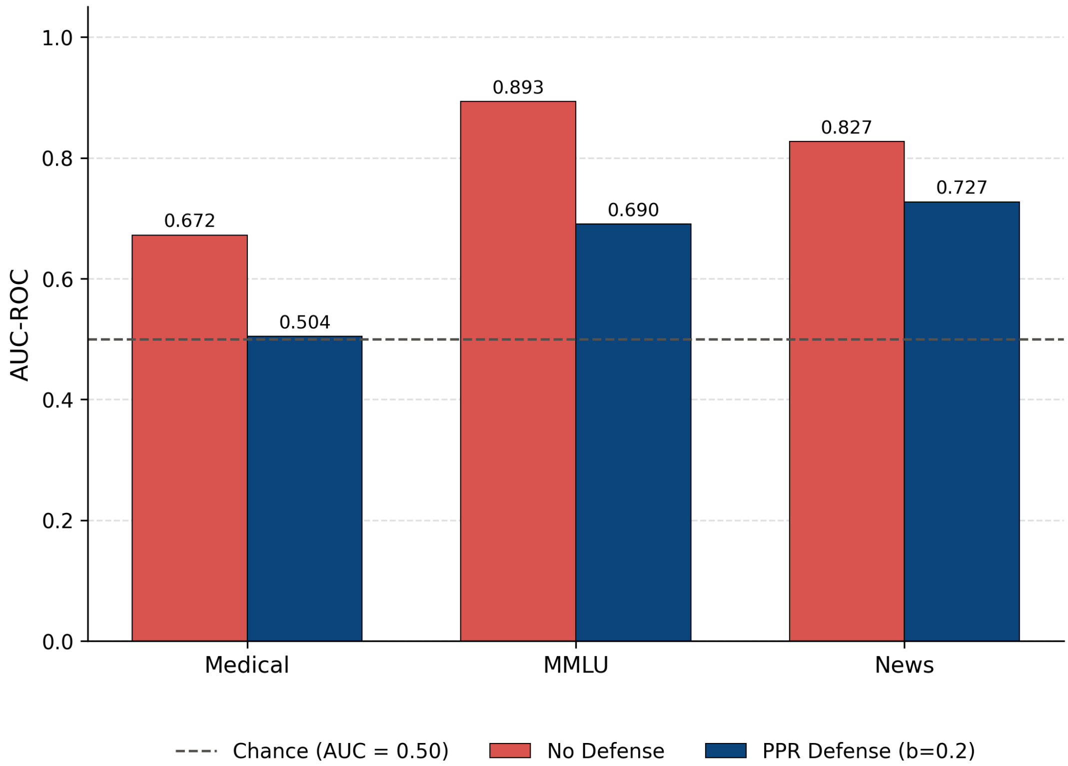

Figure 5.8: Membership inference AUC-ROC and task utility against Privacy-Preserving Retrieval noise scale (DRAG)

Table 5.31 and Figure 5.8 trace the privacy and utility axes together across five noise settings and give the privacy-utility frontier promised in Section 1.4. Both axes move monotonically. Privacy risk descends from HIGH to LOW as AUC-ROC falls from 0.8795 to 0.5152, and task F1 falls from 82.00% to 29.00% over the same range.

The frontier is not linear, which matters for choosing an operating point. Between $b = 0.05$ and $b = 0.1$ the defence buys a 0.097 reduction in AUC-ROC for 4 points of F1. The next step to $b = 0.2$ costs 30 points of F1 for a 0.178 reduction, and the step beyond that buys almost no further privacy (0.5788 to 0.5152) while removing another 22 points of utility. A deployment tuning this defense should sit at $b = 0.1$, holding the attack to MEDIUM risk while retaining 81% F1. Driving the attack to chance costs more than half the system's answer quality.

Across datasets at $b = 0.2$, Medical falls furthest, from AUC-ROC 0.6605 to 0.4771, crossing below the chance line and dropping from LOW to NEGLIGIBLE. MMLU and News start stronger and retain a MEDIUM residual after defense. Residual risk scales with how strong the underlying attack was.

5.3.5 Reputation-Based Forwarding Defense

Table 5.33: Selective Forwarding Defense Results by Strategy and Ratio (DRAG, Reputation-Based Defense)

Strategy	Ratio	Dropped Queries	Reduction	Blacklisted	Bypasses
Random	0.1	25	24.2%	1	8
Random	0.2	55	29.5%	3	23
Random	0.3	88	35.3%	6	108
Random	0.4	112	42.6%	8	155
Random	0.5	143	59.0%	9	206
High-Connectivity	0.1	30	60.5%	2	46
High-Connectivity	0.2	60	68.4%	4	130
High-Connectivity	0.3	90	68.8%	6	198
High-Connectivity	0.4	120	72.2%	8	312
High-Connectivity	0.5	150	74.6%	10	440

Note: Reduction is computed against Table 5.17. Hit rate is identical between attack-only and defended conditions at every row.

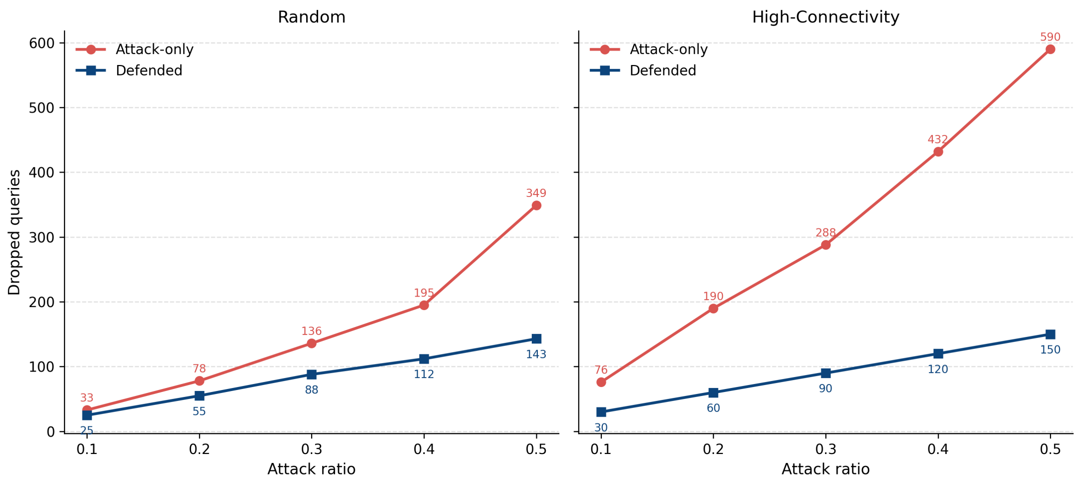

Figure 5.9: Dropped queries vs. attack ratio, attack-only vs. defended, by targeting strategy (DRAG)

Table 5.33 and Figure 5.9 show the reputation defence's benefit growing with attack severity. Under Random targeting, dropped queries fall from 33 to 25 at ratio 0.1, a 24.2% reduction, rising to 59.0% by ratio 0.5. Under High-Connectivity targeting the same pattern holds at larger scale, from 60.5% at ratio 0.1 to 74.6% at ratio 0.5. Peers blacklisted rise from 2 to 10 and bypasses from 46 to 440 over that range, so a more aggressive attacker supplies the reputation system with more evidence to act on.

Hit rate is identical between defended and undefended conditions at every ratio. Topic replication already provides enough redundant paths for the undefended router to recover within its hop budget at this network size, so the defence contributes routing efficiency rather than availability. Average hops rise marginally under defense, from 1.32 to 1.42 at Random ratio 0.4, a bounded cost for eliminating most wasted hops.

5.3.6 Layered Congestion Defense

Table 5.34: DDoS Defense Results, Mid Intensity (DRAG, All Datasets, Layered Defense)

Dataset	Strategy	QFR (%)	Avail. (%)	EM	F1	SS	Avg Hops
Medical	Random	19.17	75.0	0.6583	0.7424	0.7729	2.73
	Targeted	15.00	70.0	0.6667	0.7828	0.8177	2.24
	Sequential	12.50	65.0	0.6750	0.7813	0.8166	2.55
MMLU	Random	19.17	75.0	0.7000	0.7000	0.7349	2.18
	Targeted	15.00	70.0	0.7417	0.7417	0.7830	1.86
	Sequential	12.50	65.0	0.7750	0.7750	0.8165	1.93
News	Random	19.17	75.0	0.7083	0.7083	0.7051	2.59
	Targeted	15.00	70.0	0.7333	0.7333	0.7282	2.42
	Sequential	12.50	65.0	0.7583	0.7583	0.7539	2.49

Note: Layered defence = adaptive routing weight, circuit breaker, query hedging and load-shedding mitigation. Compare against Table 5.20.

Table 5.35: DDoS Defense Results, High Intensity (DRAG, All Datasets, Layered Defense)

Dataset	Strategy	QFR (%)	Avail. (%)	EM	F1	SS	Avg Hops
Medical	Random	21.67	30.0	0.6000	0.7013	0.7336	2.89
	Targeted	24.17	40.0	0.5833	0.6891	0.7218	3.02
	Sequential	40.83	20.0	0.4917	0.5544	0.5822	3.58
MMLU	Random	21.67	30.0	0.6750	0.6750	0.7147	2.27
	Targeted	24.17	40.0	0.6500	0.6500	0.6876	2.51
	Sequential	40.83	20.0	0.5167	0.5167	0.5363	3.25
News	Random	21.67	30.0	0.6583	0.6583	0.6525	2.68
	Targeted	24.17	40.0	0.6833	0.6750	0.6696	2.92
	Sequential	40.83	20.0	0.5167	0.5167	0.5037	3.48

Note: Compare against Table 5.21.

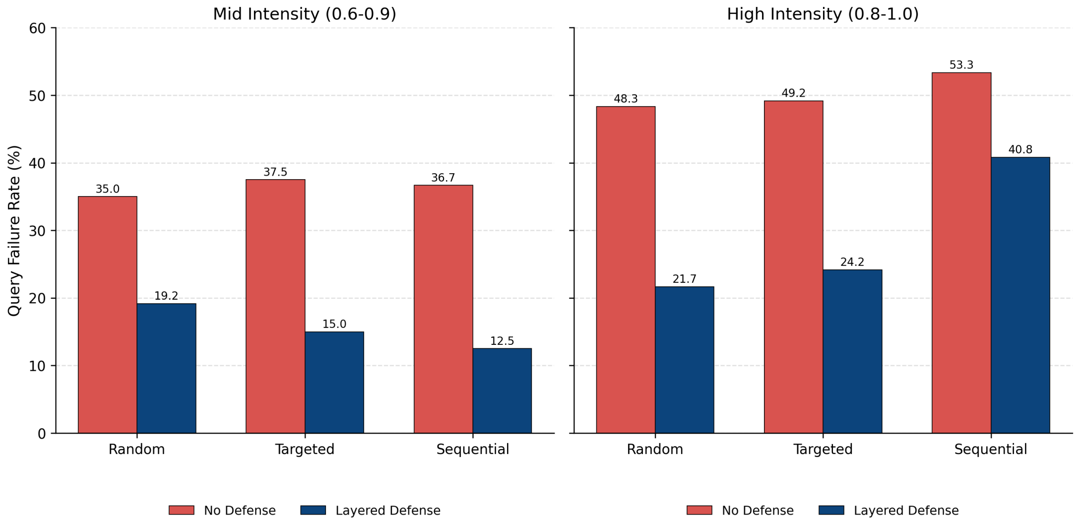

Figure 5.10: Query failure rate by strategy and intensity, with and without layered defense (DRAG)

At mid intensity, Table 5.34 cuts query failure from 35.00% to 19.17% under Random targeting and from 36.67% to 12.50% under Sequential, and lifts availability from 0–10% to 65–75%. In Table 5.35, plotted alongside the undefended condition in Figure 5.10, the recovery narrows and becomes strategy dependent. Random holds a 55.2% relative reduction (48.33% to 21.67%) while Sequential falls to 23.4% (53.33% to 40.83%) and recovers only 20.0% availability, the worst defended outcome across all six strategy and intensity combinations. A round-robin sweep leaves the circuit breaker and hedging layers the least uncongested capacity to redirect toward, so the defense erodes precisely against the strategy that spreads damage most evenly.

5.3.7 Reliable-dRAG Defenses

We scoped defense work to DRAG, which serves as the framework's demonstration of the feedback loop from evaluation to design change. Section 5.2's Reliable-dRAG results are therefore reported undefended throughout, and Section 5.6 records the resulting asymmetry as a limitation.

5.4 Statistical Analysis

5.4.1 Sample Sizes and Repetition

Every DRAG run in this chapter evaluates 100 questions. We verified the count per run from the simulator's own log line and from a record count over the saved test cases rather than from a raw line count, since several MMLU items contain embedded newlines inside quoted CSV fields. Membership inference splits its 100 items 70 members to 30 non-members. Knowledge extraction issues 5 queries per topic over 10 topics, which caps the reachable extraction rate at 50% and should be read as the ceiling for those tables rather than 100%.

Reliable-dRAG results report the mean and standard deviation over seeds 0, 42 and 123 wherever the attack permits repetition. Knowledge extraction uses 54 probes per seed per source. Data poisoning evaluates 7 questions, so each question carries 14.3 accuracy points and the standard deviations in Table 5.5 are correspondingly wide. The membership inference held-out evaluation in Table 5.16 uses 10 seeds never touched during composite tuning.

5.4.2 Sweep Design and Monotonicity

We treat monotonicity across a parameter sweep as the acceptance criterion for DRAG results, in place of the confidence intervals a seed-varied design would produce. A degradation curve that rises with attack strength, and a defence ordering that holds at every point on that curve, constitute stronger evidence than a single point with error bars, because they test the claim at several operating conditions rather than one.

Three sweeps satisfy the criterion without exception. Cross-peer validation orders above replication, which orders above no defence, in all 6 MMLU cells and all 9 cross-dataset cells. Routing manipulation degrades monotonically across 4 claim ratios, with topics hijacked tracking the ratio linearly. Privacy-preserving retrieval moves both privacy and utility monotonically across 5 noise settings.

Two sweeps contain a reversal, and we report both rather than smoothing them. High-Degree poisoning rises from 66.00% to 68.00% F1 between the 20% and 30% ratios, which Section 5.2.1 attributes to overlap between the Data-Rich and High-Degree peer sets at 30% in a 20-peer graph. News's undefended poisoning sequence puts 10% ahead of 20% in damage, which we attribute to topic sampling rather than to the poison ratio.

5.4.3 Held-Out Validation

Reliable-dRAG's membership inference composite is the one result in this chapter tuned against data rather than specified in advance, so it carries a corresponding validation requirement. We searched candidate signal weightings over the 13 seeds the project had already observed, locked the weights, then evaluated once against five seeds generated for that evaluation and never queried before it. Table 5.16 reports mean AUC-ROC 0.620 at a 95% confidence interval of [0.546, 0.695], which excludes chance. The held-out mean sits above the 0.572 the same weighting scores on the 13 search seeds, so the search did not overfit to the data it ran on. Every one of the five held-out seeds scored above 0.50 and none inverted. Similarity alone scores 0.4732, below chance, which is why the production composite zeroes it, and restoring certainty at 0.20 lowers three further fresh seeds from 0.6133 to 0.6072.

5.4.4 Degenerate Thresholds

Three tables contain rows where a classifier threshold labelled every query a member, producing TPR and FPR both equal to 1.0: DRAG's News row in Table 5.14, and Reliable-dRAG's seeds 42 and 123 in Table 5.15. Table 5.16 records the weightings we tested and rejected alongside the production composite.

5.5 Comparisons and Relationships

5.5.1 Attack Classes Across Two Architectures

Table 5.36 places each attack class beside the mechanism it exploits in each system. The framework specifies each class independently of mechanism; applying it to a new architecture means identifying which component carries the corresponding trust assumption and instantiating against that.

Table 5.36: Attack Class Instantiation and Severity Across Both Systems

Attack Class	**DRAG Mechanism**	**Reliable-dRAG Mechanism**	**DRAG Severity**	**R-dRAG Severity**
Data Poisoning	In-memory peer knowledge base	`/poison` write path into live FAISS index	Medium (30.7% F1 loss, News)	High (66.7% degradation)
Source Selection Manipulation	TARW topic advertisement	On-chain grounding-farming; orchestrator key forgery	Medium (28.0% F1 loss at claim ratio 0.9)	Low (9.0–12.0 pp accuracy drop); see note
Knowledge Extraction	Multi-hop TARW routing	Direct per-source `/query` HTTP	Medium (30% ER, 10 peers, MMLU)	Medium (41.9% ER, gray-box)
Membership Inference	Similarity plus routing composite	Decision-match on single response	High (PRS 0.786)	Low–Medium (AUC 0.620, CI [0.546, 0.695])
Selective Forwarding	TARW with topic replication	TTL-bounded breadth-first search	Negligible (hit rate 1.00 to 0.95)	Medium (hit rate 0.98 to 0.54, simulated)
Denial of Service	Wave congestion over 20 peers	Wave congestion; live HTTP flood on 3 sources	Critical (availability 100% to 0%)	Critical (F1 0.00, 90% failure)

Note: We assign severity with the rubric in Table 5.2, taking each attack at its most damaging tested configuration. Integrity and availability rows report degradation relative to that configuration's own clean baseline rather than in absolute metric points. DRAG severities are undefended. Two rows carry qualifications. The rubric scores user-visible answer quality, so it does not capture the on-chain reliability corruption that Source Selection Manipulation inflicts on Reliable-dRAG, where Grounding-Farming drives the target source's reliability score by +680 and Key-Forgery by +20,000 while answer accuracy falls by 9.0 to 12.0 points; Section 5.6 records this as a coverage gap in the rubric. The Reliable-dRAG selective forwarding figure comes from the simulated network model rather than from the live deployment.

Three relationships come out of the comparison.

Source Selection Manipulation instantiates differently in each system and exists in neither centralised RAG nor in either system's published threat model. DRAG's routing algorithm trusts a peer's advertised topic expertise, so the attack falsifies advertisements. Reliable-dRAG's reranker trusts an on-chain reliability score derived from a grounding check, so the attack farms that check with filler content that satisfies a literal substring test. We derived the second instantiation by applying the framework's mapping step to Reliable-dRAG's reliability mechanism, not by porting DRAG's attack, and the deployed on-chain defence turned out to be structurally unable to observe it: the defence validates caller identity, transaction rate, delta size and bounds, and Grounding-Farming submits no transaction that violates any of those.

Severity inverts between systems on two classes, and the architecture explains both. Selective forwarding costs DRAG almost nothing (hit rate 0.95–1.00) because topic replication leaves redundant paths inside the hop budget, while it drives Reliable-dRAG's hit rate to 0.54, because a TTL-bounded breadth-first search spends a hop on every visit to a silent peer. Denial of service reaches a worse outcome on Reliable-dRAG (F1 0.00) than on DRAG (F1 0.41) because three data sources constitute the whole retrieval backend, while 20 peers retain partial routing paths. Redundancy, not defence quality, decides both outcomes.

Membership inference is the one class where DRAG leaks more. Its composite draws 60% of its weight from retrieval similarity and 30% from routing behaviour, and a multi-hop network exposes routing behaviour that a directly-queried orchestrator does not. The signal the attack needs is a byproduct of the routing layer, so the architecture that routes more leaks more.

5.5.2 Comparison with Existing Work

Published RAG security evaluations target centralised, single-knowledge-base deployments. SafeRAG benchmarks attack surfaces against one retrieval corpus; RAGLeak and DCMI both study membership inference against a single vector store; Zeng et al. characterise extraction and leakage in a monolithic RAG pipeline. None models source selection, multi-party trust, or knowledge partitioned across independently operated participants, because those components do not exist in the architectures they evaluate.

Three of our six attack classes have no centralised analogue. Source Selection Manipulation requires more than one source to choose between. Selective Forwarding requires a participant that can be reached but decline to answer honestly. Membership inference at the partition level, revealing which source holds a record rather than whether the system holds it, requires knowledge to be partitioned in the first place. The remaining three classes exist centrally but change character under distribution: poisoning scales its attack surface with source count and rewards strategic source selection, extraction gains a topology dependence we measured directly in Table 5.9, and denial of service acquires a redundancy dimension absent from a single-server deployment.

Our extraction results also qualify a figure that recurs in this literature. Reported extraction rates in the 20–60% range come predominantly from multiple-choice corpora, and Table 5.10 shows the same attack recovering 1–6% on free-text corpora at identical topology and query budget. Extraction severity is a function of answer structure as much as of retrieval design.

5.6 Discussions

5.6.1 Interpretation

Applying the framework twice produced a result we did not anticipate: defence effectiveness rises with attack severity in three of the five DRAG defences. Cross-peer validation holds its ordering at every poison ratio and recovers most where the attack takes most. Cross-peer validation against routing manipulation improves as the claim ratio rises, because the outlier detector's quota of three times the median grows easier to clear as an attacker claims more topics. The reputation defence against selective forwarding cuts 24.2% of drops at ratio 0.1 and 74.6% at ratio 0.5, because a more aggressive attacker supplies more evidence. Measuring any of these at one attack strength would have hidden the property, which argues for parameter sweeps over single-point evaluation when assessing a defence rather than an attack.

The layered congestion defence runs the other way, eroding from a 65.9% relative reduction at mid intensity to 23.4% at high intensity under Sequential targeting. The distinction is whether a defence needs surviving capacity to work. Reputation and quorum mechanisms need evidence, which an aggressive attacker supplies. Rerouting mechanisms need somewhere to reroute to, which an aggressive attacker removes.

5.6.2 Implications for System Design

Four design conclusions follow from the measurements.

Network size resists extraction better than connectivity tuning does. Doubling peers from 10 to 20 cut extraction from 25–30% to 20%, while connectivity mattered only in the smaller network.

Replication carries a retrieval benefit independent of security. Medical's defended F1 at 10% and 20% poisoning matched or exceeded its clean baseline, so a deployment adopting replication for integrity reasons should expect answer quality to improve rather than to degrade.

Privacy-preserving retrieval has a usable operating point and an unusable one. At $b = 0.1$ the defence holds membership inference to MEDIUM while retaining 81% task F1; beyond $b = 0.2$ it buys almost no additional privacy for large utility losses.

A defence aimed at the wrong layer provides no protection regardless of its own quality. Reliable-dRAG's on-chain caps stop forged and oversized transactions, and Grounding-Farming submits neither, so a well-implemented defence sits beside an unmitigated attack. Our framework surfaced that gap by asking which signal source selection trusts rather than which transactions the defence inspects.

5.6.3 Limitations

We evaluated two systems. Both are distributed RAG deployments with independently operated sources, and they differ enough architecturally, gossip peer-to-peer routing against a blockchain-reranked orchestrator, to test whether the attack classes transfer. Two systems demonstrate portability rather than establishing generality, and we present the framework as a proof of concept on that basis.

DRAG results come from a single random seed with parameters swept, so we report monotonicity rather than confidence intervals and cannot separate small non-monotonic movements from sampling variation. The High-Degree reversal at 30% poisoning and News's 10% against 20% reversal both fall into that gap.

Reliable-dRAG's live selective forwarding validation rests on one configuration at one seed, in which every dropped query was recovered by redundant rerouting. Whether the mock model understates real resilience or that configuration is unrepresentative is unresolved, and Table 5.19 should be read as an open question rather than a result. The live DDoS evaluation likewise runs at a single seed, though its effect size and monotonic dose-response across three severity levels leave less room for a sampling explanation.

We implemented defences on DRAG only, so Section 5.3 demonstrates the feedback loop on one architecture rather than two. Reliable-dRAG severities in Table 5.36 are therefore undefended figures and are not directly comparable to a defended DRAG deployment.

Two threat models assume more than an external attacker holds. Data poisoning assumes write access to at least one source's knowledge store, an insider or compromised-participant position we treat as a worst-case upper bound; the informative variable is which source an attacker chooses rather than whether the attack succeeds. Knowledge extraction's gray-box condition pre-filters probes to guarantee a hit, measuring post-reconnaissance efficiency rather than discovery cost, which is why we report the blind condition beside it.

Query efficiency reads 100% in every undefended extraction row because the attack issues network queries with a zero confidence threshold, so any non-empty response counts as a hit. The metric is informative as a ratio against the defended rows and misleading as an absolute measure of attack precision.

Six attack classes do not exhaust the space. Sybil attacks and collusion between sources compose over the single-adversary primitives we measured rather than introducing new trust assumptions, and cross-source correlation, inferring from several sources jointly what none reveals alone, is the clearest class our framework does not yet cover.